What Makes a Song Popular on Spotify?

Spotify is one of the biggest music platforms in the world, with millions of songs available and new ones being added every day. But not all songs get the same attention — some blow up while others barely get any streams. So, what's the difference? That's the question this project tries to answer.

I'll be using a dataset from Kaggle called the Spotify Tracks Dataset. It includes thousands of songs from different genres and time periods, with audio features like danceability, energy, loudness, tempo, acousticness, and valence, along with a popularity score for each song. To make the analysis a bit richer, I also plan to create some new variables from the existing ones — for example, a ratio between energy and danceability — to see if those help explain popularity better.

The dataset has several thousand songs in total, and each row represents one track. The popularity score will be the main variable I'm trying to understand and predict throughout the project.

I'll start by cleaning the data — handling any missing values or weird entries — and then do some exploratory analysis. This means making charts like histograms, scatter plots, and a correlation heatmap to get a feel for the data and spot any obvious patterns. From there, I'll move into hypothesis testing to check whether certain features are actually linked to popularity. For example, are more energetic songs more popular? Does a higher danceability score lead to more streams?

After the statistical side, I'll build some machine learning models to try to predict popularity. I'll start with linear regression as a baseline, then try a decision tree to see if a more flexible model does better. I'll compare the results and talk about what worked and what didn't.

By the end of the project, I want to have a clear answer to which audio features matter most for popularity, and an honest look at how well these models can actually predict it.

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